Technological Institute of the Philippines
Manila Campus
College of Computer Studies
Bachelor of Science in Computer Science

Week 08 - Laboratory
CRISP-DM (Data Understanding and Data Preparation)

Big Data Analytics
CISELEC 501A-CS41S1

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CRISP-DM: Step 1 | Business Understanding

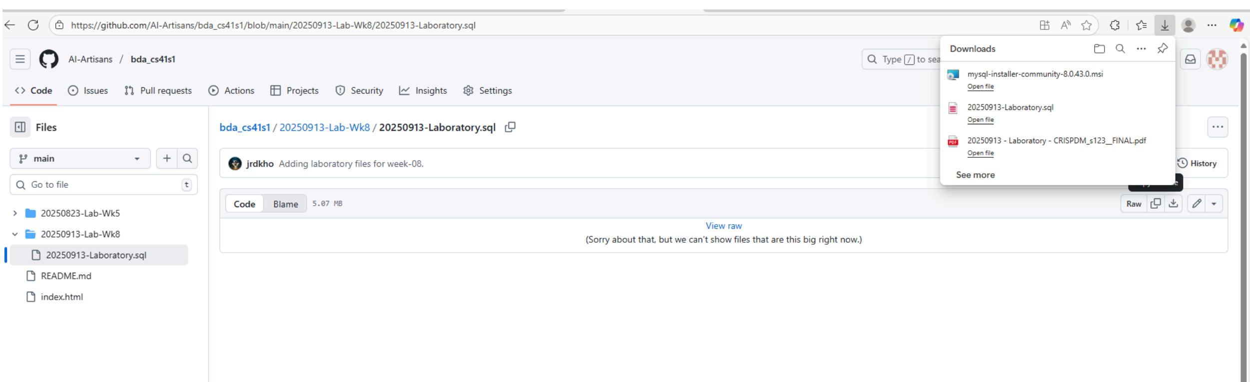
Objective:

This activity is designed to explore the initial phases of the CRISP-DM framework – Business Understanding, Data, Understanding, and Data Preparation – through scenario-based analysis. It involves identifying real world problems in the banking sector, examining a publicly available dataset, and preparing the data for future modeling. The exercise emphasizes the importance of aligning data mining efforts with clearly defined business goals, resource considerations, and contextual constraints, forming a foundation for informed decision-making and effective analytical planning.

CRISP-DM: Step 2 | Data Understanding

Instructions:

Download MySQL data-dump from GitHub.



Restore it via command line:

mysql -u [MySQL DB User] [MySQL database] < [local path of MySQL data-dump file]

Explore the table contents.

Reflection:

1.) What challenges did you experience during the MySQL data-dump restoration? (5 pts)

I initially had issues because my queries were using the wrong table name (bank instead of bank_marketing), so nothing was showing up. After fixing the table name, everything worked.

With regards to this activity, I used DB Fiddle which is an online SQL playground and I am unable to load every single data from the .sql file so i used only those that is needed on the activity

2.) With respect to the newly created MySQL table, complete the details below (5 pts):
Details of MySQL Table:
Number of Attributes:

Query #2 Execution time: 1.57ms

num_attributes
17

Number of Objects: (Run SELECT COUNT(*) FROM bank_marketing; – you should get around 65535 rows if you loaded the entire file)

Query #3 Execution time: 0.45ms

num_objects
4

3.) With respect to the objects inside the table, complete the details below (10 pts):

Results

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Query #3 Execution time: 0.45ms

num_objects
4

Query #4 Execution time: 0.96ms

Field	Type	Null	Key	Default	Extra
age	int	YES		null	
job	varchar(50)	YES		null	
marital	varchar(20)	YES		null	
education	varchar(30)	YES		null	

Results

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marital	varchar(20)	YES		null	
education	varchar(30)	YES		null	
defaulted	varchar(10)	YES		null	
balance	decimal(10,2)	YES		null	
housing	varchar(10)	YES		null	
loan	varchar(10)	YES		null	
contact	varchar(20)	YES		null	
day	int	YES		null	

day	int	YES		null	
month	varchar(10)	YES		null	
duration	int	YES		null	
campaign	int	YES		null	
pdays	int	YES		null	
previous	int	YES		null	
poutcome	varchar(20)	YES		null	
y	varchar(10)	YES		null	

CRISP-DM: Step-3 | Data Preparation

INSTRUCTIONS:

1.) Perform initial data exploration using the restored data-dump. Complete item #1 in Reflection section.

2.) Based on the summary produced in Reflection item #1, kindly perform data clean-up on the following attributes:
a. Job
b. Marital
c. Education
d. Contact

3.) Perform another set of data exploration after the data clean-up and complete item #2 in the Reflection section.

REFLECTION:
1.) Provide the initial summary for the following attributes (10 pts):
a. Job:
There are multiple categories such as management, technician, blue-collar, admin., services, retired, entrepreneur, self-employed, student, housemaid, unemployed, and several unknown values. Some inconsistencies exist, such as Adm vs admin..
b. Marital:
Contains categories like married, single, divorced, and some entries with inconsistent casing like Div.
c. Education:
Values include primary, secondary, tertiary, but also have unknown values. Some may use slightly different spellings or formats.
d. Contact:
Mostly contains values such as cellular, telephone, and a large number of unknown entries.

2.) Provide the final summary for the following attributes (15 pts):
a. Job:
All job names standardized (e.g., admin. and Adm merged into admin), unknown values either removed or replaced with none.
b. Marital:
All values converted to lowercase, Div replaced with divorced.
c. Education:
Unified categories, removed duplicates, and replaced unknown with none for consistency.
d. Contact:
Replaced unknown with none, leaving only valid contact types (cellular, telephone, none).

3.) Based on the results produced in items #1 and #2, what issues did you encounter during the clean-up process? (5 pts)
I faced issues deciding whether to delete rows with unknown values or keep them with replacements. Some job titles had abbreviations that needed manual checking before grouping. I had to ensure I did not accidentally lose valid data while cleaning.

4.) What are the possible repercussions if data clean-up is neglected during the “Data Preparation” phase of the CRISP-DM framework? (10 pts)

If data cleaning is skipped, the model may learn from incorrect or inconsistent data, resulting in inaccurate predictions. It can produce biased results, lower model accuracy, and lead to bad business decisions. It might also make analysis harder since there will be duplicates, missing values, and noise in the dataset.

Results

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Query #5 Execution time: 0.18ms

job
unemployed
services
management
technician

Query #6 Execution time: 0.11ms

marital
married
single

Query #7 Execution time: 0.1ms

education
primary
secondary
tertiary

Query #8 Execution time: 0.1ms

contact
cellular
unknown